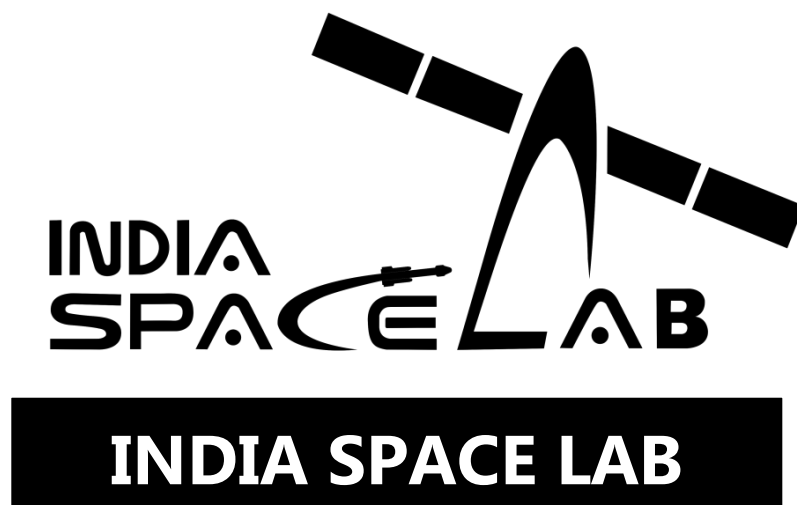


ADVANCE DRONE TECHNOLOGY
INTRODUCTION TO GUIDANCE AND CONTROL USING
PYTHON SIMULATIONS



Project Report
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1. Introduction

This project report details the comprehensive simulation and analysis of guidance and control systems for autonomous vehicles, specifically focusing on quadcopter drones and marine surface vessels. The study was conducted as part of the India Space Lab internship program, utilizing Python-based simulations within a Google Colab environment. The primary objective is to model real-world vehicle dynamics and evaluate the efficacy of feedback control systems in maintaining stable trajectories under unpredictable environmental disturbances.

2. Objectives

- To understand the fundamentals of Guidance, Navigation, and Control (GNC) for autonomous moving systems.
- To apply PID (Proportional-Integral-Derivative) controller tuning to achieve desired system states.
- To observe and mathematically mitigate the effects of environmental disturbances, such as wind shear and water currents.
- To develop advanced programmatic simulations using Python to trace complex parametric trajectories.

3. Task 1: Drone Altitude Control

3.1 System Overview

The first task involved designing a PID controller to manage the altitude of a quadcopter drone. The controller regulates the simulated vertical thrust to achieve and maintain a target altitude of 10 meters while rejecting external disturbances. The control law governing the system is:

$$u(t) = K_p * e(t) + K_i * \int e(t) dt + K_d * de(t)/dt$$

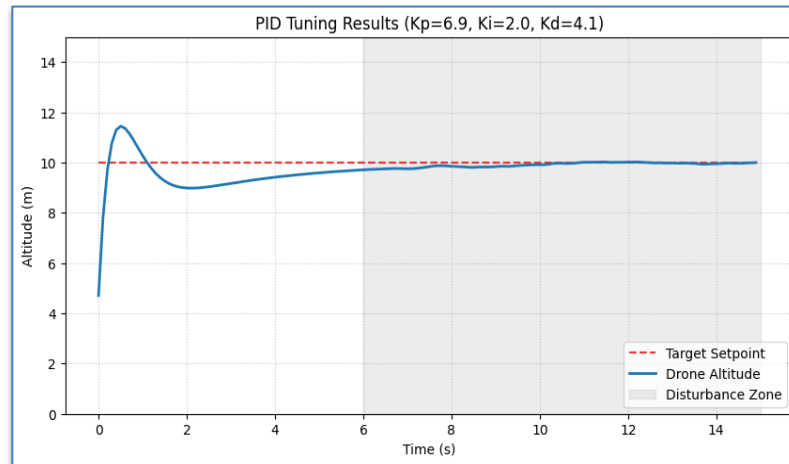
3.2 Tuning Methodology & Results

The tuning process followed a systematic approach to balance response time and system stability. Based on the simulation results, the optimal tuned parameters were **Kp = 6.9, Ki = 2.0, and Kd = 2.4**.

- **Proportional Gain (Kp = 6.9):** Provided a highly aggressive initial thrust, resulting in a fast rise time to reach the 10m target altitude quickly.
- **Derivative Gain (Kd = 2.4):** Acted as a necessary dampener (brake) against the high proportional gain. This prevented the drone from overshooting and oscillating wildly, allowing it to settle smoothly on the target line.
- **Integral Gain (Ki = 2.0):** Tuned specifically to address steady-state error introduced by environmental factors.

3.3 Disturbance Rejection

At exactly $t = 6$ seconds in the simulation, a constant downward wind disturbance was injected. The graph demonstrates that while the wind attempts to force the drone downward, the Integral term (K_i) successfully accumulates the error and increases thrust, forcing the drone back to the precise 10m target altitude.



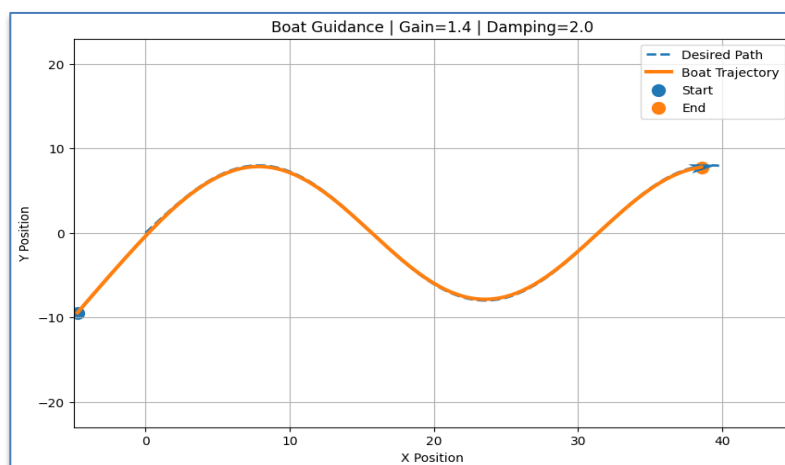
4. Task 2: Autonomous Boat Guidance

4.1 Path Tracking Dynamics

The second task shifted focus to horizontal plane guidance for an autonomous surface vessel (boat). The objective was to track a predefined sinusoidal trajectory using a feedback correction mechanism based on heading and cross-track errors.

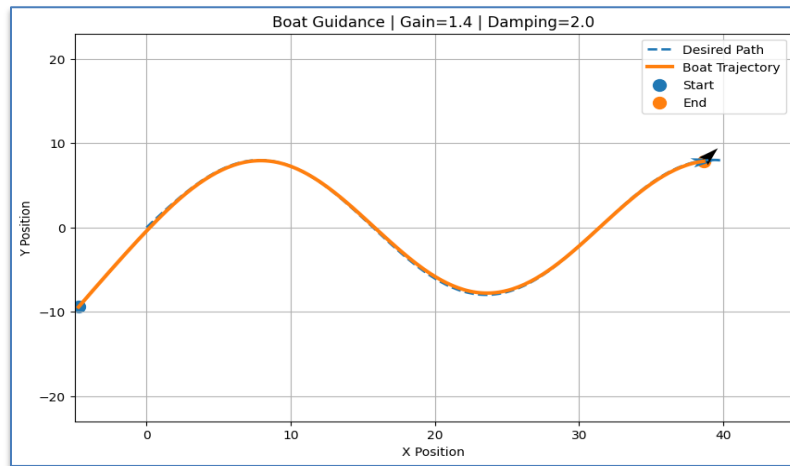
4.2 Ideal Conditions (Without Disturbance)

First, the boat was simulated under ideal environmental conditions, with the lateral and longitudinal water currents set to zero ($\text{current}_x = 0.0$, $\text{current}_y = 0.0$). The steering gains were tuned, allowing the boat to perfectly overlay its actual trajectory onto the desired path.



4.3 Handling Environmental Currents (With Disturbance)

To test robustness, strong environmental currents were introduced ($\text{current}_x > 0$, $\text{current}_y > 0$). These forces continuously attempt to push the vessel off its intended course. The heading and cross-track error gains were adjusted, enabling the vessel to autonomously angle (crab) its heading into the current, successfully maintaining its position on the desired path despite the continuous external force.



5. Bonus Task: Figure-Eight Trajectory Tracking & Control

5.1 Mathematical Modeling

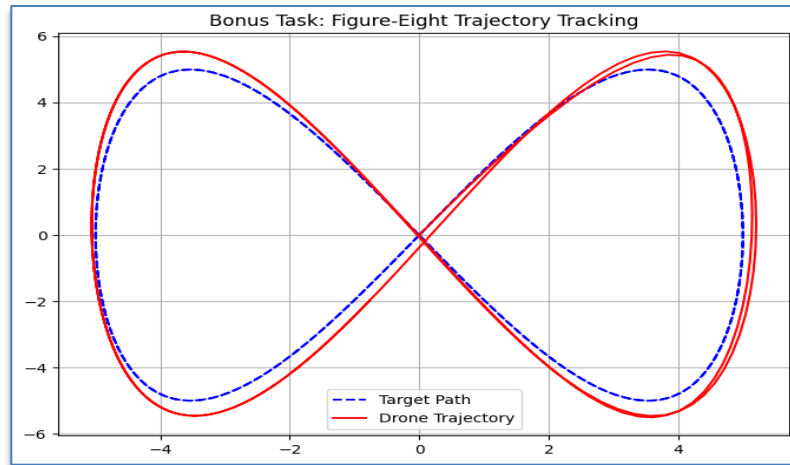
To demonstrate advanced path-planning capabilities, a continuous figure-eight trajectory—mathematically defined as the *Lemniscate of Bernoulli*—was simulated. The parametric equations governing the target waypoints are:

- $x_{\text{target}}(t) = A * \sin(\omega t)$
- $y_{\text{target}}(t) = A * \sin(2\omega t)$

5.2 Control Architecture & Disturbance

A Dual-Axis Proportional-Derivative (PD) control loop was engineered to track the moving coordinate frame. To evaluate the controller's robust stability, an unmodeled external disturbance (representing a sudden, constant lateral wind shear) was programmatically injected at $t > 5.0$ seconds.

The simulation proves that the PD controller successfully compensated for the drift, forcing the drone to aggressively crab into the wind while maintaining strict adherence to the Lemniscate curve. This demonstrates high control loop stability and tracking fluidity.



6. Conclusion

This project successfully demonstrated the core principles of autonomous guidance and control. Through hands-on Python simulations, the nuances of PID tuning were deeply explored, highlighting the critical balance between response speed, stability, and disturbance rejection. The ability to mathematically compensate for unpredictable environmental factors—whether wind shear on a drone or marine currents on a boat—is validated as an essential framework for the reliable and safe operation of modern autonomous vehicles.